

Development of Digital Data Collection Tools

(Google Forms)

Data Collection and Management Module



Aim

To equip participants with the practical skills necessary to design, customize, and deploy high-quality Google Forms for data collection, training, and research.

Learning Outcomes

By the end of this training, the participants should be able to:

1. Understand the fundamental core features and functionalities of Google Forms.
2. Design structured and ethical surveys.
3. Customize and optimize forms.
4. Deploy and manage forms effectively.
5. Collaborate with team members in the design of surveys.

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INTRODUCTION

A google form is a versatile, free web-based tool for gathering information, feedback and data.

1.1 Uses of Google Forms

Google forms are used for a wide range of settings. For example;

A) In Education:

- **Quizzes and Tests:** Create multiple-choice, short answer or fill-in-the blank questions to assess student learning.
- **Homework Submission:** Students can upload assignments directly to the form.
- **Attendance Tracking:** Easily record student attendance or participation.
- **Behavior Reports:** Document and track incidents or student behaviour.
- **Parent Surveys:** Collect feedback from parents on school activities or classroom environment.

B) Business:

- **Customer Feedback:** Gather opinions and reviews about products or services.
- **Order Forms:** Accept orders for products or services with customizable fields.
- **Event Registration:** Manage RSVPs and collect attendee information.
- **Job Applications:** Create structured forms for applicants to submit resumes and details.
- **Internal Surveys:** Poll employees on workplace satisfaction or gather input on company decisions.

C) Personal or Community Use:

- **Party Planning:** Coordinate dates, food preferences and guest lists.
- **Volunteer Sign-ups:** Organize community service or religious events.
- **Polls and Voting:** Let people vote on group decisions like trip destinations or club activities.
- **Family Coordination:** Find the best time for reunions or vacations

1.2 Advantages of using Google Forms

Some of the advantages of using google forms are:

1. **Free to Use:** Google forms are accessible to anyone with a Google account at no cost.
2. **Ease of Use:** Google Forms have a simple and intuitive interface, making it accessible to users of all skill levels.
3. **Easy Collaboration:** Multiple people can edit and manage the form.
4. **Automatic Real-Time Data Collection:** Responses are stored in google sheets for analysis.
5. **Custom Themes:** Ability to personalize a google form with preferred color and image.
6. **File Uploads:** Respondents can attach documents and/or images.

1.3 Disadvantages of using Google Forms

Some of the disadvantages of using google forms are:

1. **Customization Limited Design:** Google forms have a basic layout. You can't fully customize the look and feel of the form to match branding.
2. **No Native Payment Integration:** You can't collect payments directly via the form. You will require 3rd party tools or manual follow-up.
3. **Google Account Requirement:** Features such as file uploads or some sharing settings require respondents to sign in using their google accounts. This can act as a barrier to non google users.

4. **Basic Reporting Tools:** No advanced visualizations or dashboards.
5. **Limited Question Types:** Google forms have less variety of questions compared to other survey-type platforms.
6. **Performance Issues:** Uploading/Downloading speeds may slow down when many users access the form simultaneously.
7. **No Offline Access:** Forms must always be filled out when respondents are online.
8. **Security and Privacy Concerns:** Google forms are not ideal for collecting sensitive or confidential information without additional security measures.

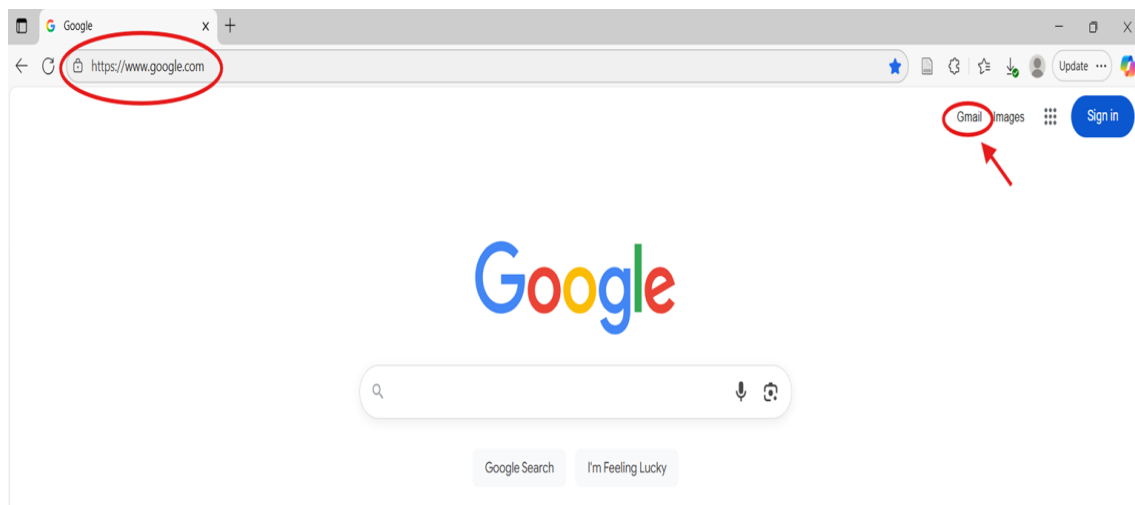
GOOGLE FORM CREATION

2.1 Setting up a Google Drive Account

Google Drive is a free, web-based office suite and data storage service offered by google. It allows users to create online documents and edit them collaboratively. Google drive offers the form option that can be used to generate online surveys. To set-up an online questionnaire (google form), you need a **gmail** account.

2.1.1 Setting up a Gmail Account

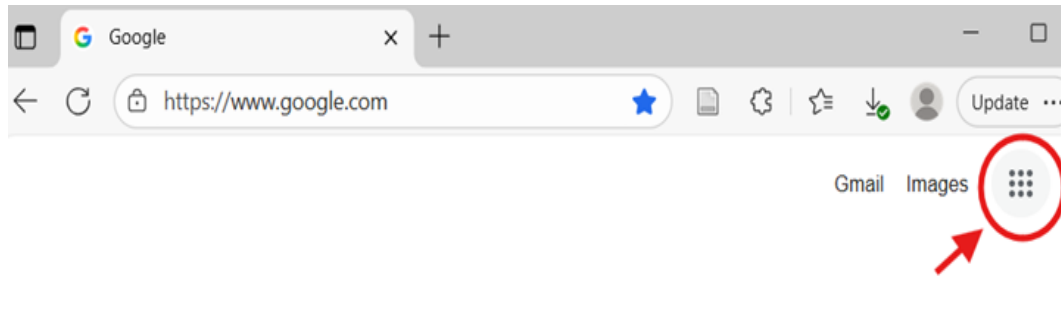
1. Go to www.google.com and click on [gmail](mailto:).



2. Click on “**Create an Account**” and fill in your required set-up details.
3. You can always access your gmail account by going to [mail.google.com](mailto:) or to www.gmail.com

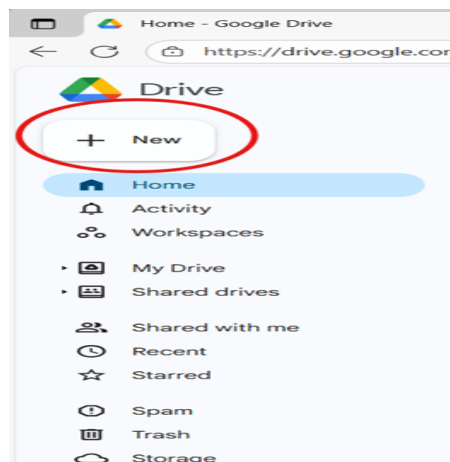
2.1.2 Accessing your Google Drive Account

1. Go to **www.google.com** and ensure you are signed in to your gmail account.
2. Click on the **dots** and select **Drive**.

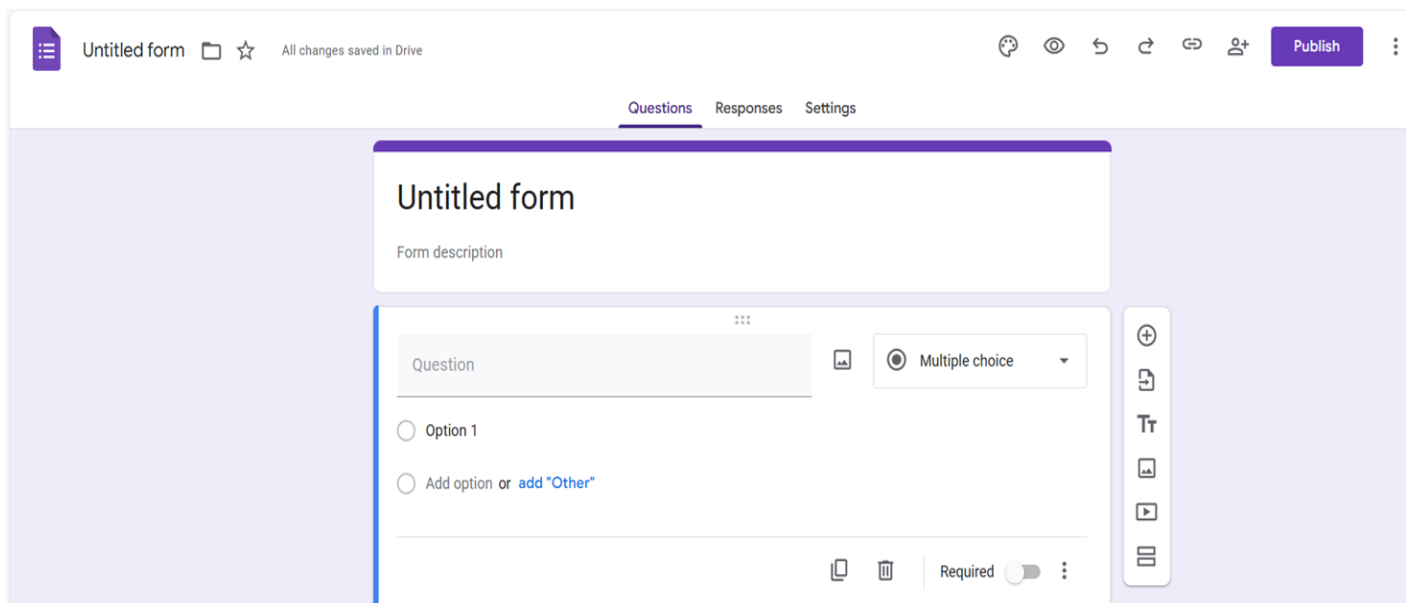


2.2 Creating your first Google Form

1. On your Google Drive Folder, Click on the **New** Button.



2. Select **Google Forms** then **Blank form**. The *untitled form* will appear.



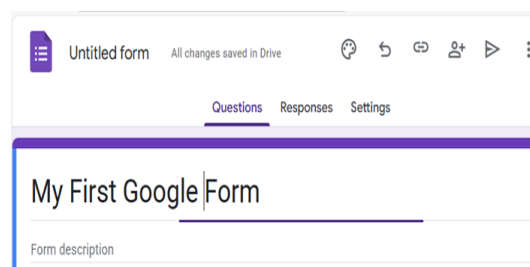
2.2.1 Components of a Google Form

Google forms have several key components:

1. **Title and Description:** Gives the form's name and an optional description to explain its purpose.
2. **Questions:** This is the heart of the form—can be a short answer, multiple choice, checkboxes, etc.
3. **Sections:** Used to divide the form into multiple pages or parts for better organization.
4. **Images and Videos:** Media elements that can be embedded to enhance questions or provide context.
5. **Response Validation:** Rules to ensure correct input (e.g., number ranges, text format).
6. **Required Toggle:** Ensures respondents must answer a question before submitting the form.
7. **Theme and Design:** Customization options for colors, fonts, and background images.
8. **Submit Button:** Final button that respondents click to send their answers.
9. **Confirmation Message:** Custom message shown after submission (e.g., “Thanks for your response!”).
10. **Settings Menu:** Controls form behavior—like collecting emails, limiting responses, or quiz mode.

2.2.2 Form Title

The title gives the form's name. To edit the title of your form, replace the text in the *Untitled form* box.



2.2.3 Form Description

If you have any additional guidance for the recipients, the instructions may be typed on the *Form Description* section.

Facts about Mosquitoes

This survey aims to understand community knowledge, experiences, and practices related to mosquitoes and mosquito-borne diseases.

- This survey should only be filled for Capacity Building Purposes Only.
- You should be 18 years and older for eligibility.

GOOGLE FORM QUESTIONS

A Google Form question is a prompt or item that is included in a Google Form to collect specific information from respondents. Each question can be customized based on the type of response you want.

3.1 Google Form Question Features

Google form offers additional features to its question section to enhance usability, data quality, and efficiency.



- **Add Question:** Insert a new question of any type.
- **Import Question:** Reuse standardized questions that you have created across multiple forms.
- **Title and Description:** Add headings and explanatory text to section or groups of questions.
- **Image Insertion:** Add images to questions or sections.
- **Video Embedding:** Embed youtube videos for context or instruction.
- **Section:** Divide the form into multiple parts. This feature is useful for long forms or conditional logic.
- **Duplicate Question:** Quickly copy a question to reuse its format.

- **Required Toggle:** Toggling on this option makes the question to be mandatory. The feature ensures critical data is collected.
- **Delete Question:** remove a question or section from form.

3.2 Components of a Google Form Questions

Google Form questions are made up of several key components that help respondents gather and organize their responses effectively. The key components are:

1. Question Title
2. Question Type
3. Answer Options
4. Response Controls
5. Description (Optional)
6. Image or Video (Optional)

The image shows a Google Form question editor interface with several components highlighted by red boxes and numbered 1 through 8:

- 1:** Question Title (Text input field)
- 2:** Description (Text input field)
- 3:** Question Type (Dropdown menu showing "Short answer")
- 4:** Answer Options (Text input field for "Short answer text")
- 5:** Image or Video (Icon for adding media)
- 6:** Required Toggle (Toggle switch labeled "Required")
- 7:** Copy Icon (Icon for copying the question)
- 8:** Delete Icon (Icon for deleting the question)

A dropdown menu is open from the bottom right, showing options: "Show", "✓ Description 2", and "Response validation 6".

3.2.1 Question Title

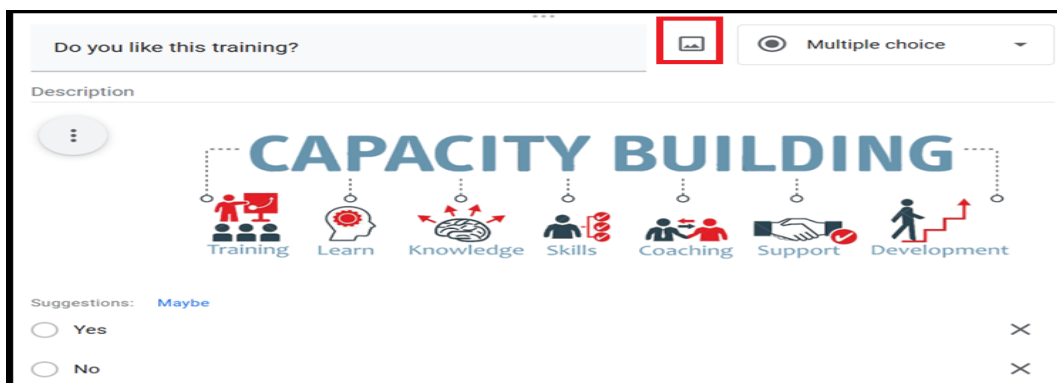
This is the main prompt or query of the question you are asking. It should be clear and concise



A screenshot of a Google Form editor showing a question titled "What is your name?". The title is highlighted with a red rectangular box. To the right of the title is a "Multiple choice" dropdown menu. Below the title are two radio button options: "Option 1" and "Add option or add 'Other'". At the bottom right, there are icons for duplicating and deleting the question, a "Required" toggle switch, and a vertical ellipsis menu.

3.2.2 Inline Question Images (Optional)

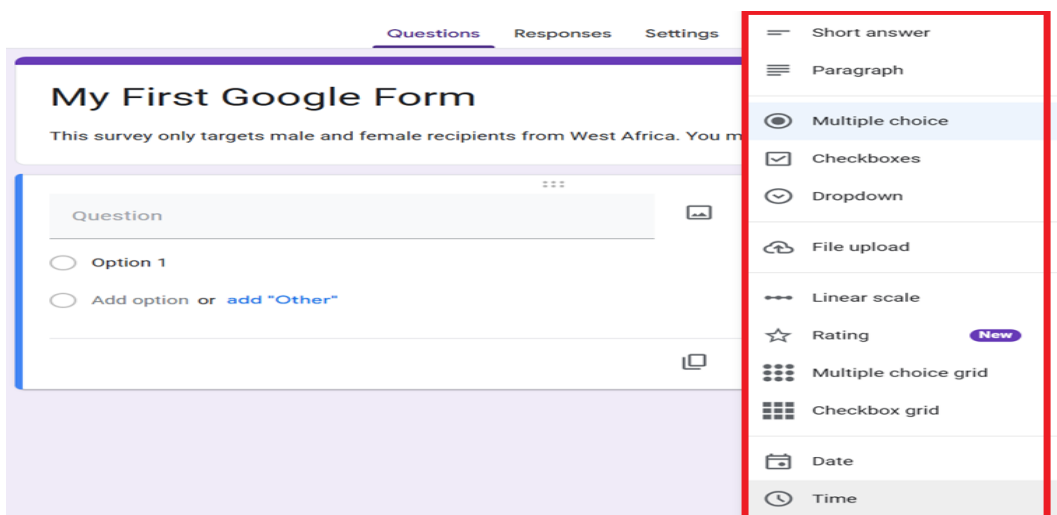
Google forms allow you to insert images to support questions or enhance visual appeal.



A screenshot of a Google Form editor showing a question titled "Do you like this training?". To the right of the title is a red box highlighting the image icon. Below the title is a "Multiple choice" dropdown menu. The description area contains a large graphic titled "CAPACITY BUILDING" with icons for Training, Learn, Knowledge, Skills, Coaching, Support, and Development. Below the graphic are two radio button options: "Yes" and "No". At the bottom right, there are icons for duplicating and deleting the question.

3.2.3 Question Types

Google Forms provide a layout for different types of questions based on their usage and functionality. Question types determine how respondents answer different questions.

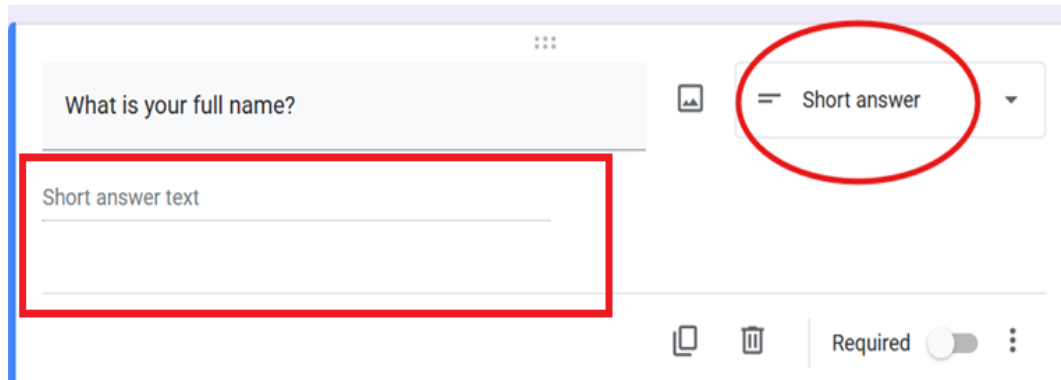


A screenshot of the Google Forms editor showing the "Questions" tab. The question type selection menu is open, displaying a list of question types: Short answer, Paragraph, Multiple choice (selected), Checkboxes, Dropdown, File upload, Linear scale, Rating (marked as "New"), Multiple choice grid, Checkbox grid, Date, and Time. The menu is highlighted with a red rectangular box.

3.2.4 Answer Options

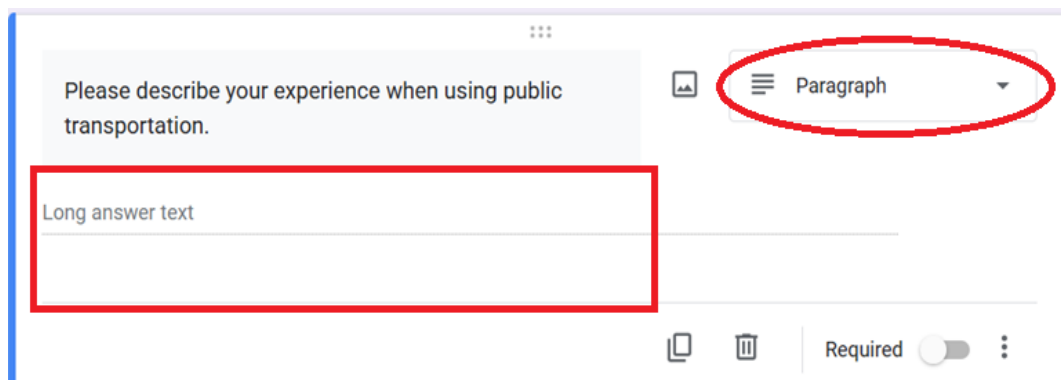
These are the types of responses or choices that the respondent will be allowed to fill in.

- **Short answer:** Brief text responses such as names, email addresses or one-sentence answers.



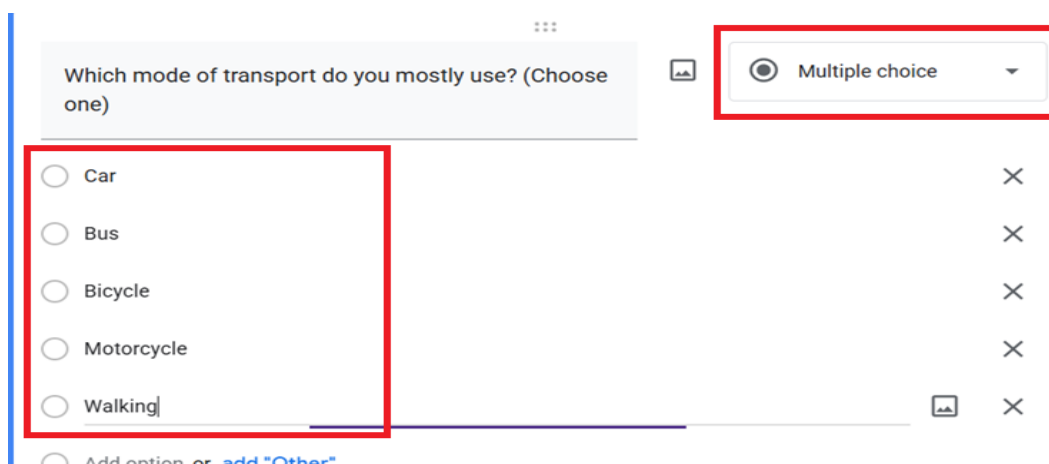
The screenshot shows a survey question: "What is your full name?". To the right of the question is a dropdown menu set to "Short answer", which is circled in red. Below the question is a text input field labeled "Short answer text", also outlined in red. At the bottom right, there are icons for copying and deleting, a "Required" toggle switch, and a three-dot menu.

- **Paragraph:** Longer text responses such as feedback, explanations or descriptions



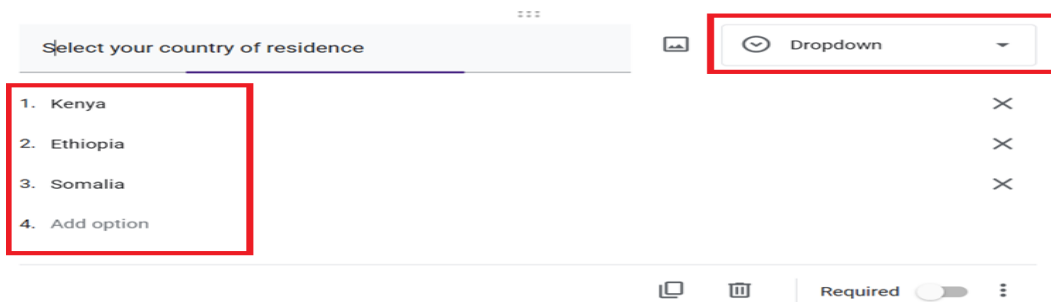
The screenshot shows a survey question: "Please describe your experience when using public transportation.". To the right of the question is a dropdown menu set to "Paragraph", which is circled in red. Below the question is a large text input field labeled "Long answer text", also outlined in red. At the bottom right, there are icons for copying and deleting, a "Required" toggle switch, and a three-dot menu.

- **Multiple Choice:** Respondents select only one option from such a list.



The screenshot shows a survey question: "Which mode of transport do you mostly use? (Choose one)". To the right of the question is a dropdown menu set to "Multiple choice", which is outlined in red. Below the question is a list of radio button options: "Car", "Bus", "Bicycle", "Motorcycle", and "Walking". Each option has a red 'X' to its right. The list is enclosed in a red box. At the bottom, there is a link to "Add option or add 'Other'" and a three-dot menu.

- **Dropdown:** Similar to multiple choice but options appear in a compact drop down menu.

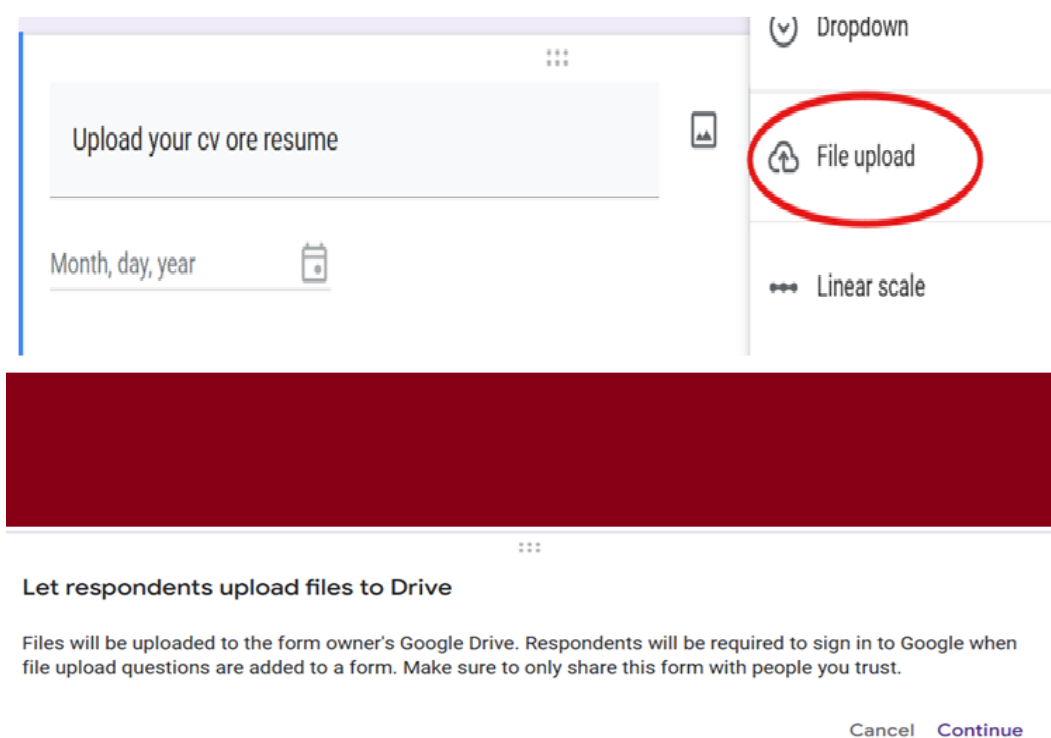


Select your country of residence

1. Kenya
2. Ethiopia
3. Somalia
4. Add option

Required

- **File upload:** Allows respondents to attach files such as documents, images e.t.c. Option requires respondents to sign in.



Upload your cv ore resume

Month, day, year

File upload

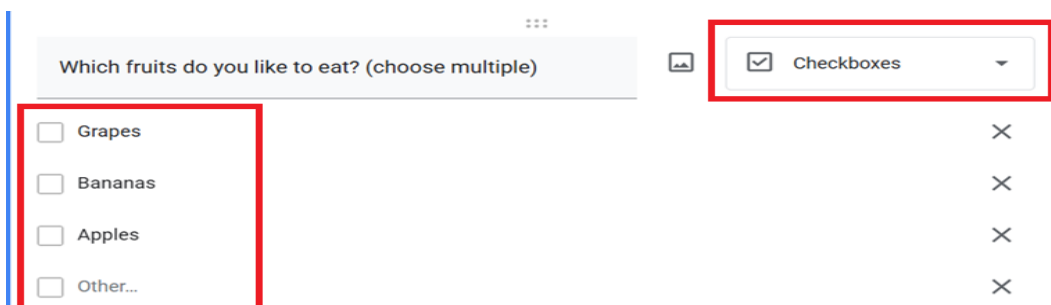
Linear scale

Let respondents upload files to Drive

Files will be uploaded to the form owner's Google Drive. Respondents will be required to sign in to Google when file upload questions are added to a form. Make sure to only share this form with people you trust.

Cancel Continue

- **Checkboxes:** Allows selection of multiple options. Useful where more than one answer applies.



Which fruits do you like to eat? (choose multiple)

☐ Grapes

☐ Bananas

☐ Apples

☐ Other...

Required

- **Linear Scale:** Let users rate something on a scale. Ideal for feedback on satisfaction or agreement levels.

How satisfied are you with the training?

Linear scale

1 to 5

1 Not satisfied

5 Very satisfied

- **Rating:** Similar to Linear Scale but users rate using the star-ratings.

Rate our services

Rating

5

1 2 3 4 5

☆ ☆ ☆ ☆ ☆

- **Multiple choice grid:** Users choose one option per row across several categories.

Rate your understanding in the following subjects

Multiple choice grid

Rows	Columns
1. Mathematics	☐ Poor
2. English	☐ Fair
3. Science	☐ Good
4. Add row	☐ Excellent
	☐ Add column

- **Checkbox grid:** Users can select multiple options per row. Useful for scheduling or preference matrices.

Select your most preferred meeting times

Checkbox grid

Rows	Columns
1. Monday	☐ Morning
2. Tuesday	☐ Afternoon
3. Wednesday	☐ Evening
4. Add row	☐ Add column

- **Date:** Collects a specific date such as birthdate, appointment e.t.c.

Enter your date of birth

Month, day, year

Date

- **Time:** Collects a specific time such as meeting time, duration e.t.c.

What time do you usually wake up?

Time

Time

3.2.5 Response Validation

This option gives the set rules for acceptable input e.g number ranges, email format e.t.c.

What is your age (years)?

Short answer

Short answer text

Number Greater than 17

Custom error text

Required

Show

Description

✓ Response validation

3.2.6 Question Description (Optional)

What is your full name?

Short answer

Indicate your surname first in capital letters

Short answer text

Required

Show

✓ Description

Response validation

Please describe our experience when using public transport

Long answer text

3.3 Google Form Sections

Sections add new pages to the google form. To add a new section to a form, click on the section feature indicated below.



3.3.1 Importance of Sectioning Forms

Sectioning forms:

1. Improve form organization and flow by breaking long forms into manageable parts. This helps respondents focus on one topic at a time.
2. Incorporates conditional logic or branching by allowing respondents “***Go to section based on answer***”. This tailors form experience based on responses enabling respondents to skip irrelevant sections.
3. Enhances clarity and structure allowing users to add titles and descriptions in each section.
4. Facilitates grouping of related data for easier analysis.

3.3.2 Practical Assignment 1

Structure a google form with the following five sections and the following questions:

- **Section 1: Consenting Statement**

Dear Participant. Please not by continuing with this survey, you confirm that you understand its purpose and voluntarily agree to participate. Your responses will be kept confidential and used for research purposes only. Do you consent to this?

Conditional Logic: If **Yes**; Proceed with survey, If **No**; ***Submit Form***

- **Section 2: Participant Information**

- What is your full name?
- Select your country of residence
- Upload your cv or resume.
- Enter your date of birth
- What is your age (years)?

- **Section 3: Random Questions**

- Please describe your experience when using public transportation.
- Which mode of transport do you mostly use?
- Which fruits do you like to eat? (choose multiple)

- **Section 4: General Feedback**

- How satisfied are you with the training?
- Rate our services.
- Rate your understanding in the following subjects
- Select your most preferred meeting times.
- What time do you wake up?

- **Section 5: Thank you for your response**

Include General Statement: Thank you for taking the time to complete this survey. Your insights are incredibly valuable and will contribute meaningfully to our efforts to improve and inform future initiatives. We deeply appreciate your willingness to share your experiences and perspectives—your input helps drive positive change.

GOOGLE FORM SETTINGS MENU

The Settings menu in Google Forms allows users to control how their form behave for respondents and how responses are managed. This menu controls form behavior—like collecting emails, limiting responses, or quiz mode. It's divided into several key sections that enhance functionality, privacy, and customization.

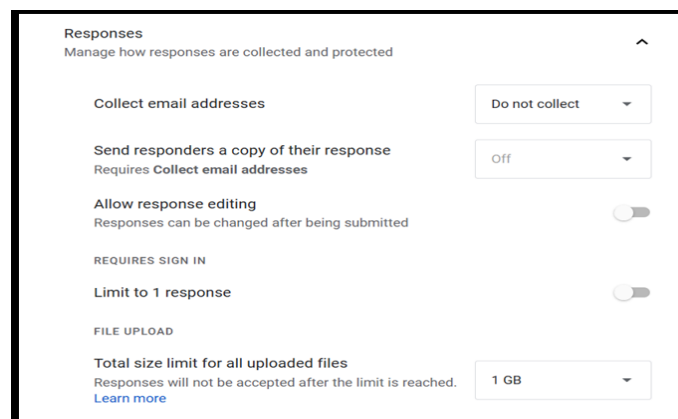


Google Form Settings menu is subdivided into three (3) main categories:

- General (Response)
- Presentation
- Quiz Mode

4.1 General (Response) Settings

The general settings manage how responses are collected and protected.

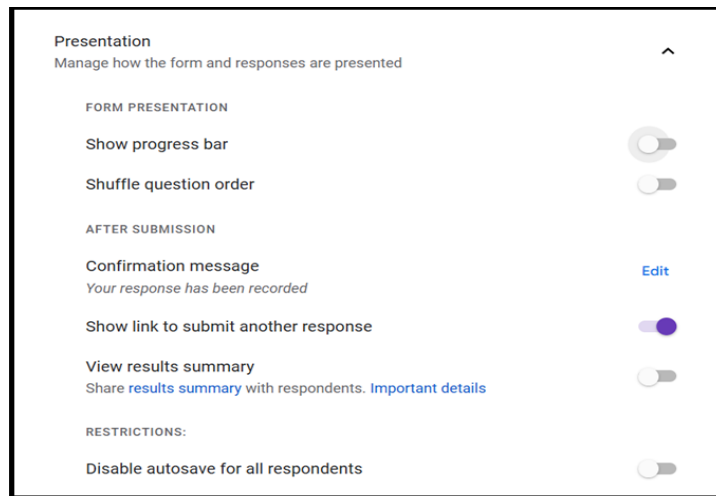


It's features include:

1. **Collect email addresses:** Useful for follow-up or authentication.
2. **Limit to 1 response:** Prevents duplicate entries (requires sign-in).
3. **Allow response editing:** Allows respondents to revise their answers after submission.

4.2 Presentation Settings

The presentation settings manages how the form and responses are presented.



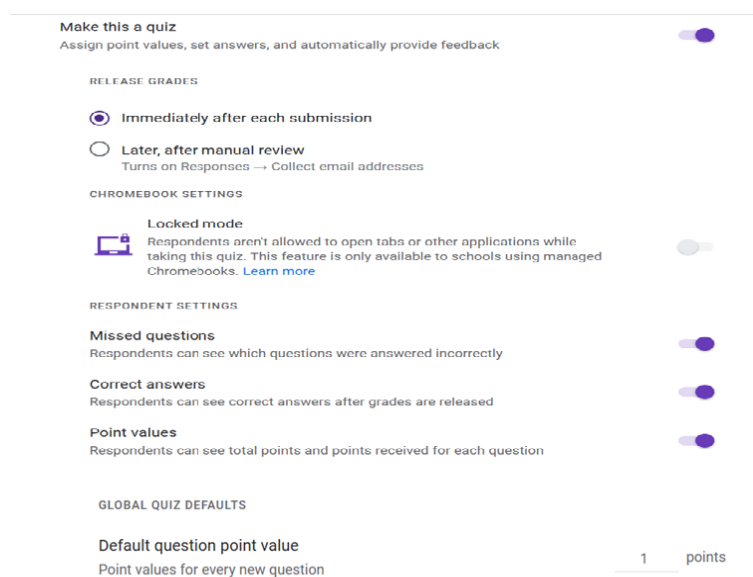
The screenshot shows the 'Presentation' settings panel. At the top, it says 'Presentation' and 'Manage how the form and responses are presented'. Below this, there are several sections: 'FORM PRESENTATION' with 'Show progress bar' (toggle on) and 'Shuffle question order' (toggle off); 'AFTER SUBMISSION' with 'Confirmation message' (text: 'Your response has been recorded', with an 'Edit' link), 'Show link to submit another response' (toggle on), and 'View results summary' (toggle off, with text: 'Share results summary with respondents. Important details'); and 'RESTRICTIONS:' with 'Disable autosave for all respondents' (toggle off).

It's features include:

1. **Progress bar:** Shows how far along the respondent is to submission of form.
2. **Shuffle question order:** Randomizes questions to reduce bias or cheating.
3. **Confirmation message:** Custom thank-you or next-step instructions after submission.
4. **View results summary:** See summary charts and text responses. Lets respondents view aggregated results.

4.3 Quiz Mode Settings

Make this a quiz setting converts the form questions to allow grading. The setting allows users to assign values, set answers and automatically provide feedback.



The screenshot shows the 'Quiz Mode' settings panel. At the top, it says 'Make this a quiz' and 'Assign point values, set answers, and automatically provide feedback' (toggle on). Below this, there are several sections: 'RELEASE GRADES' with 'Immediately after each submission' (selected) and 'Later, after manual review' (radio button, with text: 'Turns on Responses → Collect email addresses'); 'CHROMEBOOK SETTINGS' with 'Locked mode' (toggle on, with text: 'Respondents aren't allowed to open tabs or other applications while taking this quiz. This feature is only available to schools using managed Chromebooks. Learn more'); 'RESPONDENT SETTINGS' with 'Missed questions' (toggle on, text: 'Respondents can see which questions were answered incorrectly'), 'Correct answers' (toggle on, text: 'Respondents can see correct answers after grades are released'), and 'Point values' (toggle on, text: 'Respondents can see total points and points received for each question'); and 'GLOBAL QUIZ DEFAULTS' with 'Default question point value' (text: 'Point values for every new question', value: '1 points').

4.3.1 Practical Assignment 2

Create a quiz with the following questions, options and solutions:

1. Which parasite causes malaria? (Answer: B. Plasmodium)

- A. Trypanosoma
- B. Plasmodium
- C. Giardia
- D. Leishmania

2. What virus causes yellow fever? (Answer: A. Flavivirus)

- A. Flavivirus
- B. Retrovirus
- C. Coronavirus
- D. Rhinovirus

3. Which mosquito-borne disease can cause birth defects? (Answer: B. Zika)

- A. Dengue
- B. Zika
- C. Malaria
- D. Chikungunya

4. What is the best prevention method for malaria?

(Answer: C. Mosquito nets and repellents)

- A. Antibiotics
- B. Vaccines
- C. Mosquito nets and repellents
- D. Vitamin supplements

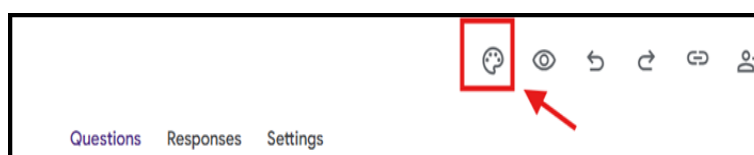
5. Which mosquito species transmits West Nile Virus?

(Answer: C. Culex)

- A. Aedes
- B. Anopheles
- C. Culex
- D. Mansonia

THEME CUSTOMIZATION IN GOOGLE FORMS

Theme customization enables users to reinforce their branding, improve readability, and make the form more engaging. It allows users to change colors, fonts, and background images to match your organization or topic. To customize the theme, click on the color palette icon below:



Google form theme customization setting has 4 major features:

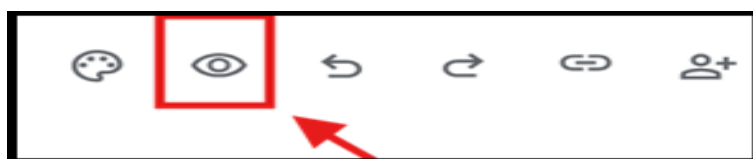
1. **Header Image:** Upload a banner or choose from Google's gallery. This section is ideal for logos or course titles.
2. **Theme Color:** Sets the accent color for buttons and highlights. Users should choose a color that aligns with their organization or topic.
3. **Background color:** Lightens or darkens the form's backdrop. Users are advised to stick to soft tones for readability.
4. **Font Style:** Choose from Basic, Decorative, Formal, or Playful. Users are advised to use *Formal* or *Basic* for professional surveys.



GOOGLE FORM PREVIEW MODE AND PUBLICATION

6.1 Preview Mode

Google forms allow users to open a live preview of the form as respondents will see it. To get a preview of your form, click on the eye icon highlighted below.



Previewing a form before sharing is crucial for ensuring a smooth and accurate user experience. It allows users to identify any errors in question design, formatting, or logic before respondents encounter them; preventing frustration and incorrect responses. This step is extremely important instead of editing forms that have already collected responses.

While previewing forms, you can test:

- The navigation transition between sections.
- The conditional logic (e.g., “Go to section based on answer”)
- The Required questions and response validation
- Overall form layout and flow

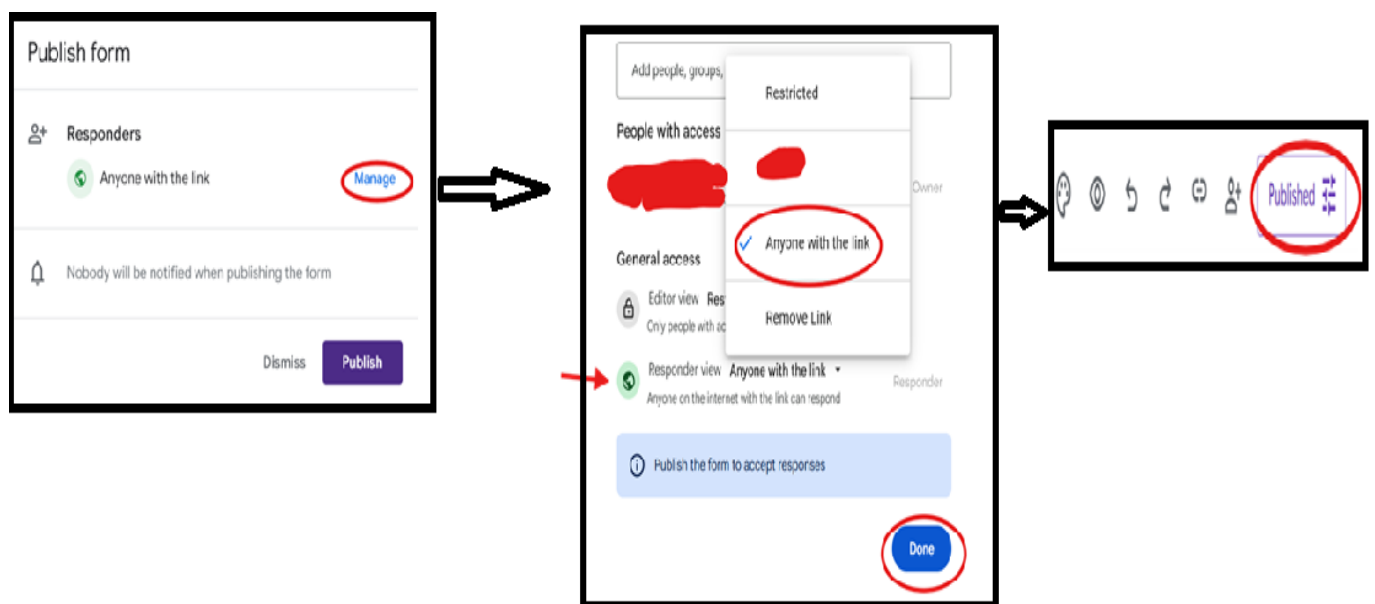
Remark: It is recommended that users use the form preview mode to test their form before sharing it publicly or with trainees.

6.2 Form Publishing

Publishing a Google Form is essential as it makes the form accessible to the intended audience for collecting responses. Unpublished forms appear blank to responders, even with the correct link. To publish a google form, click on the **Publish** button.



Publishing allows you to control who can access and respond to your form, either through specific individuals, a general link, or within a specific organization. To specify and give restrictions on who responds to your published google form; Click on **Publish** then **Manage** and select the drop down menu under **Responder view**.

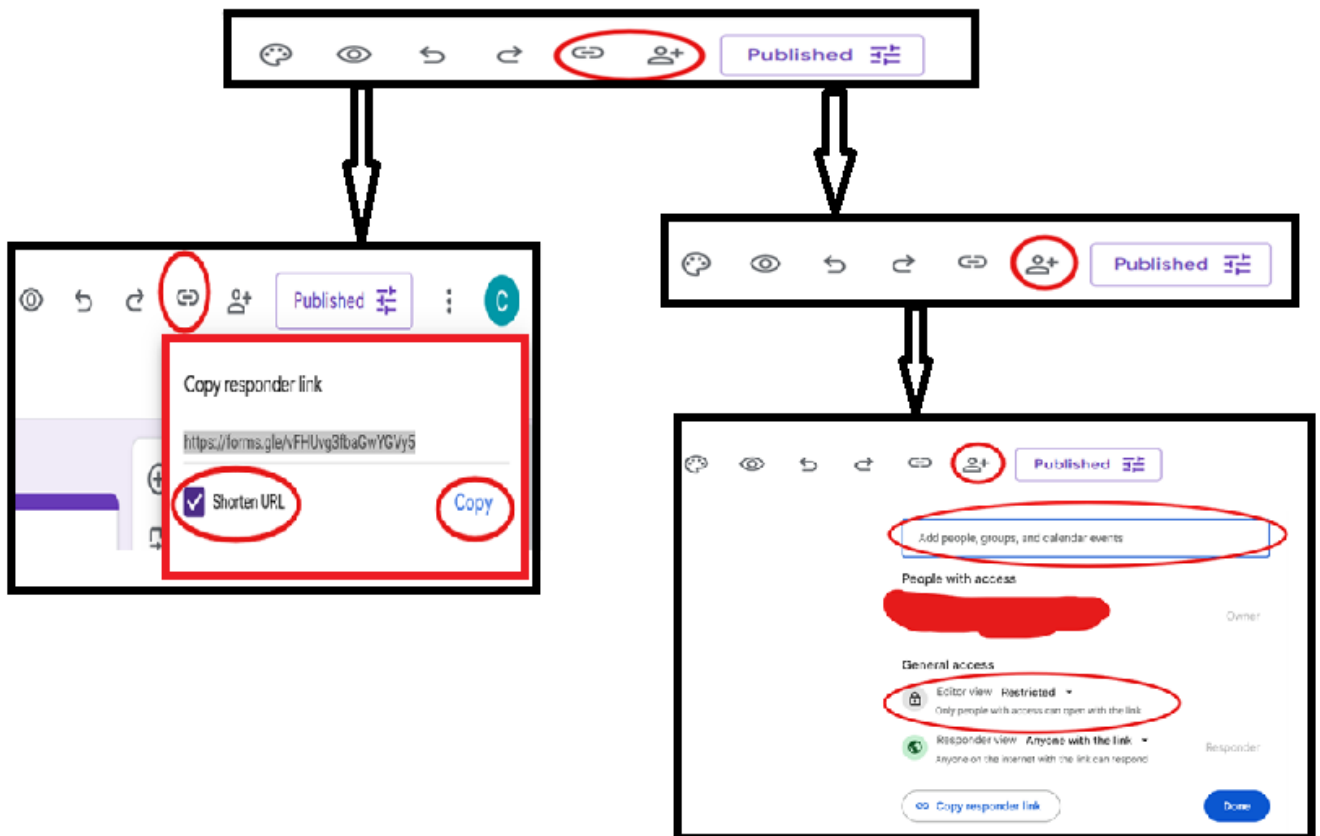


Other users can collaborate on editing your google form components by giving them access through the **Editor view**.

To publish the form, click **Done** and the form will be online ready to receive responses. A user can confirm the publication status of their form once it changes from **Publish** to **Published**.

6.3 Sharing a Published Google Form

There are several methods for sharing a published google form through sending it via email, sharing a link, embedding it on a website, or sharing it on social media. When sharing the form, you can choose to allow others to collaborate on the form or simply only respond to it.



GOOGLE FORM RESPONSES

Google Form response is the recorded set of answers provided by a user who fills out and submits a Google Form.

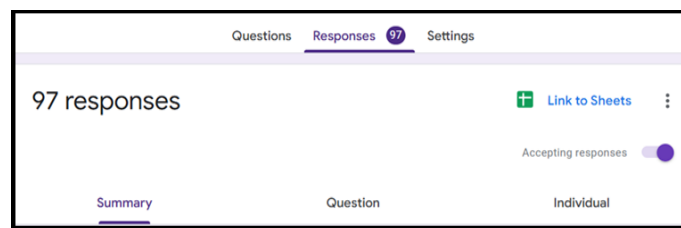


The key features of google form responses are:

- **Automatic Collection:** Responses are saved instantly once the form is submitted.
- **Response Summary:** Google Forms generates charts and graphs for quick visual analysis.
- **Individual View:** You can view each respondent's answers one at a time.
- **Export to Google Sheets:** Responses can be linked to a spreadsheet for deeper analysis and record-keeping.
- **Timestamped Entries:** Each response includes the date and time of submission.
- **Editable Settings:** You can allow respondents to edit their responses after submission (optional).

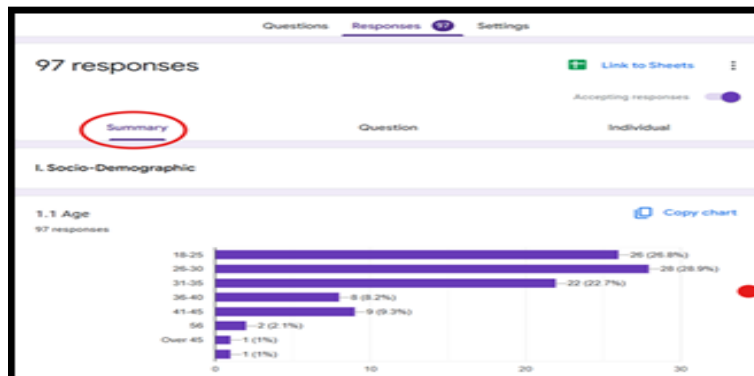
7.1 Viewing Responses

Viewing responses has 3 types of views: Summary, Question and Individual.



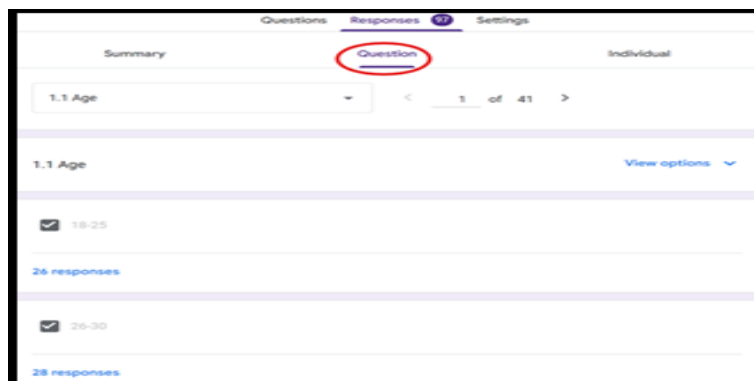
7.1.1 Summary View

Provides a summary of all responses, including graphical representations for certain question types (e.g., multiple choice, checkboxes).



7.1.2 Question View

A sub-tab used to view how all respondents answered a specific question. It allows you to navigate between questions.



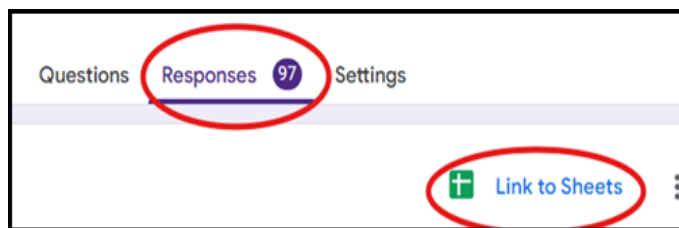
7.1.3 Individual View

A sub-tab used to see how each individual respondent completed the entire form. Allows you to switch between individual responses.

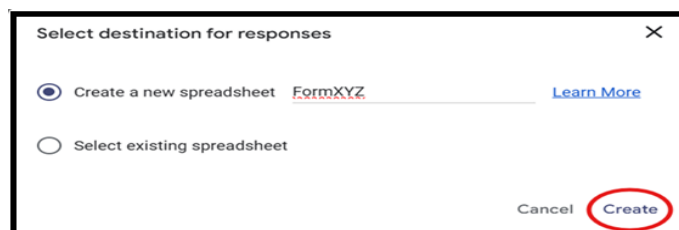
Option	Selected
18-25	☐
26-30	☐
31-35	☐
36-40	☒
41-45	☐
Other:	

7.2 Linking to Google Sheets

To link a google form to a google sheet, In the “Responses” tab, click the green Sheets icon.



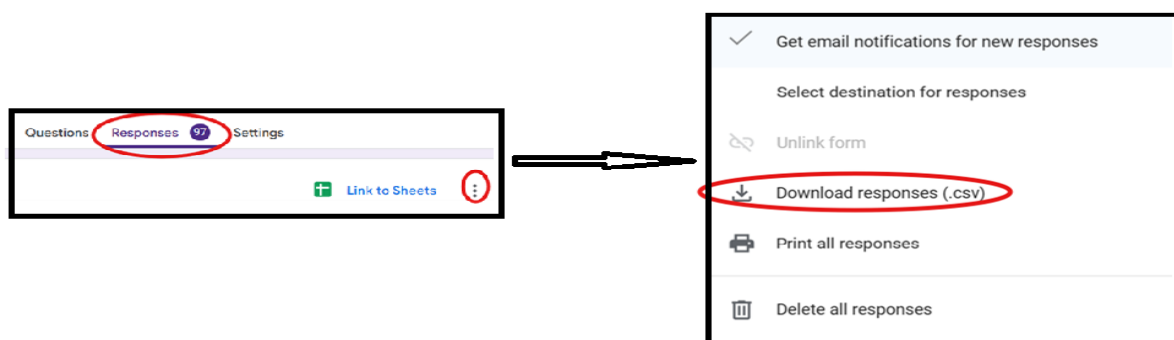
This creates a linked Google Sheet with all responses. A user can choose to create a new spreadsheet or link to an existing one for data analysis or record keeping.



NB: Each row has one respondent and each column has one question.

7.3 Exporting Data to Excel (.csv Format)

Google forms allow users to download responses in .csv format. This allows for online use or integration with other tools for analysis. To download google form data in .csv format; From the “Responses” tab click on the three dots



7.4 Response Management

For ethical and responsible data handling:

- Turn off accepting responses when your survey closes.
- Use pseudonyms or codes instead of names if anonymity is needed.
- Store Sheets in a secure Google Drive folder with restricted access.
- Consider adding a consent question in the form to document approval.